A Case Competition on Public Finance & Economic Growth

FISCONOMICS

REVIVING TONK DISTRICT: A FISCAL TRANSFORMATION CHALLENGE


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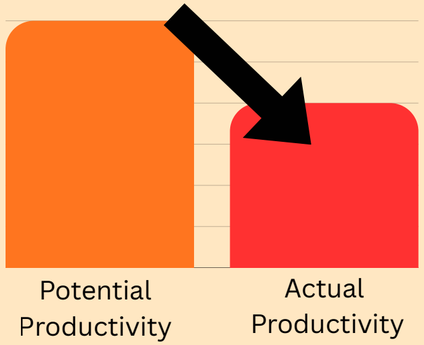
Team Name: TONKABELL

Institute: IIT MADRAS

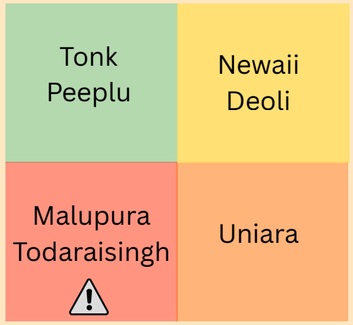
Diagnostic | Reallocation Proposal | Economic Impact | Roadmap

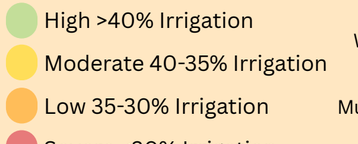
Rain Led Farming

65% of agricultural activity is rain dependent despite it contributing 42.2% of its GDDP.

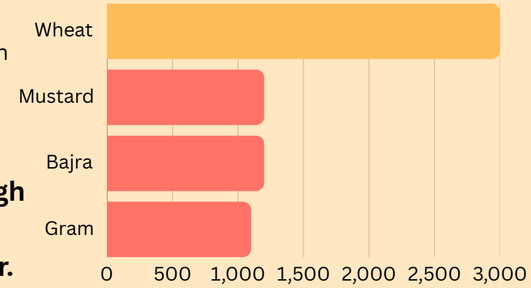


Irrigation deficit suppresses **20%** agricultural productivity resulting is Rs.720 crore annual loss

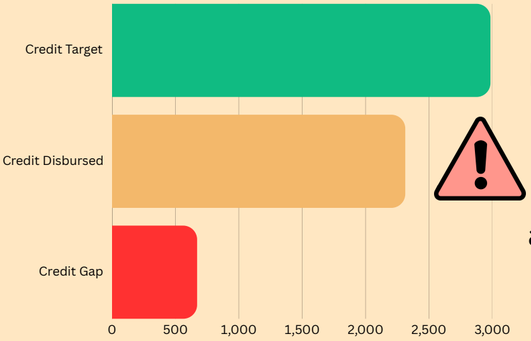




Malpura and Todaraisingh form Tonk's primary irrigation deficit cluster.



Rain-fed crops yield ~60% less than irrigated wheat, signaling water-driven productivity constraints.



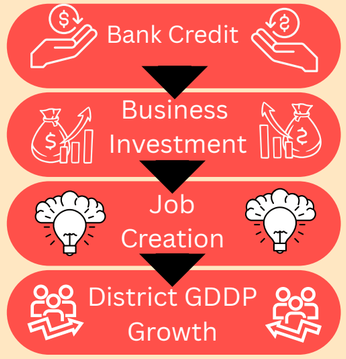
Nearly 22.5% of planned district credit —**about ₹671 Cr**— remains undelivered, limiting capital available for economic activity and investment.

Infrastructure Constraints
Irrigation covers only 34.5% of farmland, leaving ~65% agriculture rain-dependent.

Institutional Constraints
Despite 27.6% MGNREGA spending on water assets, irrigation coverage remains low at 34.5%.

Irrigation gaps stem from weak infrastructure and inefficient water asset utilization.

Credit gaps exceed 20% across all productive sectors, constraining both agriculture and industry.

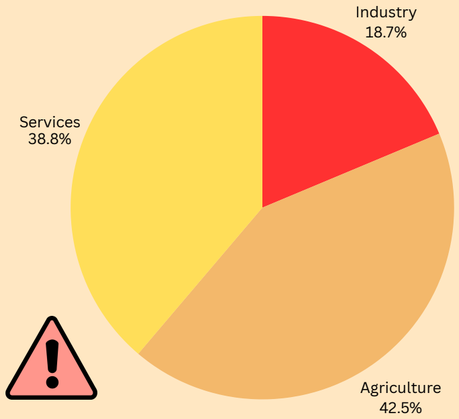


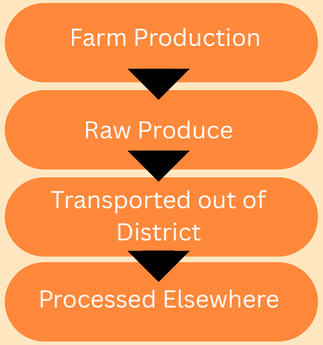
Without credit, the entire economic growth chain weakens.

78.7%

banking coverage — 45 branch deficit limits formal credit access.

Industry contributes <20% of the economy, far below growth districts where industry contributes 25–30%.





Tonk exports raw produce but imports processed value.

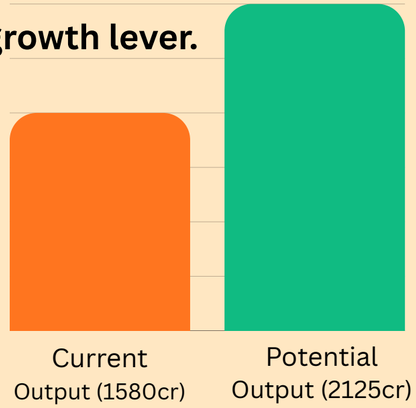
Industrial expansion is the district's largest untapped growth lever.

Finance
MSME **credit gap stands at 25.4%**, limiting capital for enterprise expansion.

Infrastructure
Industry contributes only 18.6% of GDDP, reflecting weak manufacturing infrastructure.

Ecosystem
Most agricultural output leaves Tonk unprocessed, causing significant value leakage.

Industrial growth is constrained by financing gaps, weak infrastructure, and limited local value addition



Weak industrialization limits value capture, suppressing ~₹900 Cr in potential economic output.

₹806 Cr Allocation: Three High-Multiplier Sectors

Irrigation & Agro-Processing MSMEsRoads & Credit, respectively

Irrigation	Agro-Processing MSMEs	Roads & Credit
₹280 Cr · 34.7%	₹320 Cr · 39.7%	₹206 Cr · 25.6%

 Irrigation & Agri	₹280 Cr	 Agro-Processing MSMEs	₹320 Cr	 Roads & Credit Infra	₹206 Cr
FM: 1.98× (construction) · Rationale: 42.8% of GDDP rain-dependent		FM: 1.98× (construction) · Rationale: 42.8% of GDDP rain-dependent		FM: 1.98× (construction) · Rationale: 42.8% of GDDP rain-dependent	
• Micro-irrigation (drip/sprinkler), 3 lakh Ha	₹150 Cr	• 3 agro-processing clusters: mustard oil, groundnut, maize starch (Tonk/Newai/Malpura)	₹180 Cr	• Rural market roads: 60–70 km linking Todaraisingh & Malpura to mandis & clusters (PMGSY convergence)	₹140 Cr
• Check dams & water harvesting (MGNREGA convergence)	₹70 Cr	• Cold storage network: 10 units × 5,000 MT capacity	₹60 Cr	• Rural haats & market yards in 3 underserved blocks	₹40 Cr
• Crop diversification: shift to mustard (₹5,650 MSP) & groundnut (₹6,377 MSP)	₹35 Cr	• ITI skill centres at cluster sites (12,000 workers)	₹40 Cr	• Business Correspondent expansion: 500 agents targeting 45-branch gap	₹26 Cr
• Agri extension, soil health, FPO formation	₹25 Cr	• MSME credit guarantee fund (unlocks ₹200+ Cr bank lending at 5× leverage)	₹40 Cr		

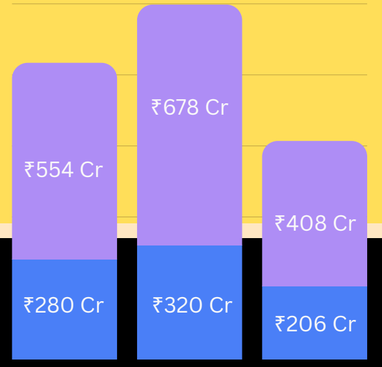
Why not education, health, or rural development?

- These sectors already receive ₹412 Cr, ₹288 Cr, and ₹457 Cr respectively, the **three largest allocations**.
- They are adequately funded relative to their **short-term GDDP multiplier**.
- Education has a **15–20 year** income gestation period.

The ₹806 Cr must go where the multiplier is highest and the structural gap is widest.

GDDP Impact Projection

₹806 Cr input → ₹1,641 Cr GDDP impact



Irrigation unlocks 42.8% of GDDP trapped behind 34.5% coverage; Agro-Processing deploys the district's highest multiplier (2.12×) against a ₹144 Cr MSME credit gap; Roads & Credit closes the 45-branch void blocking ₹671 Cr in stalled lending.

Diagnostic → Reallocation Proposal → Economic Impact → Roadmap

SECTOR	INVESTMENT (Cr)	MULTIPLIER	GDDP IMPACT (Cr)
Agriculture & Irrigation	280	1.65	462
MSME / Manufacturing	320	2.12	678
Infrastructure / Roads	206	1.98	408

Total economic impact close to 1641 Cr

KEY ASSUMPTIONS

- Sector multipliers used from case data
- Impact realised over 5 years period
- Productivity increases from irrigation and infrastructure investments

Rain-dependent Economy

↓

Irrigated Agriculture

↓

Local Manufacturing

↓

Integrated District Market

Improved road connectivity lowers transport costs, expands market access, strengthens supply chains, and supports the creation of ~**60,000 jobs** across MSMEs and allied sectors.

The projected GDDP gains translate into three structural improvements in Tonk's economy

MSME / Manufacturing Expansion

Expanded MSME clusters increase manufacturing output, create employment and supplier networks, and enable greater local value addition through in-district processing of agricultural produce.

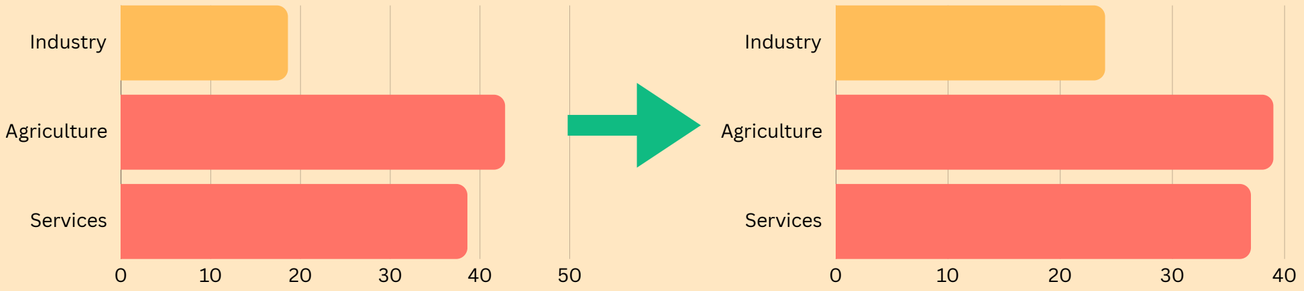
Agricultural Productivity Gains

Expanded irrigation reduces rainfall dependence, boosts crop yields and farmer incomes, and increases agricultural output that feeds local agro-processing industries.

Market & Trade Efficiency

Improved road connectivity lowers transport costs, accelerates market access, and strengthens supply chains, enabling MSMEs to scale production and trade activity.

Every ₹1 invested produces ~₹2.04 in district GDP.



MSME expansion increases industry's share of Tonk's economy, shifting the district toward a more diversified production structure.

806 Cr strategic reallocation across agriculture, MSMEs, and roads generates nearly double the economic value, creates thousands of jobs, and structurally improves rural market access.

Diagnostic → Reallocation Proposal → Economic Impact → Roadmap



₹806 CRORE GROWTH OVER THE YEARS

Year 1



District infrastructure investment plan

- Socio economic baseline **survey** across blocks, **DPC approval**
- Irrigation and infrastructure **gap mapping**
- Identify **priority blocks**(Malpura and Todaraisingh)
- Setup **district monitoring** dashboard

Year 2



Core infrastructure projects initiated

- **Irrigation** canals and water facility assets
- **Road connectivity** to farm clusters
- **Land allocation** for MSME clusters

Year 3



Productive infrastructure operational

- MSME clusters operational
- **Agro processing units** launched
- **Villages to mandis** connectivity
- Digital asset **tracking** system

Year 4



Local enterprises and farmer incomes begin scaling

- **Credit linkage** for farmers & MSMEs
- **Skill training** for agro processing, manufacturing & services
- FPO led supply chain aggregation

Year 5



District-wide economic growth momentum

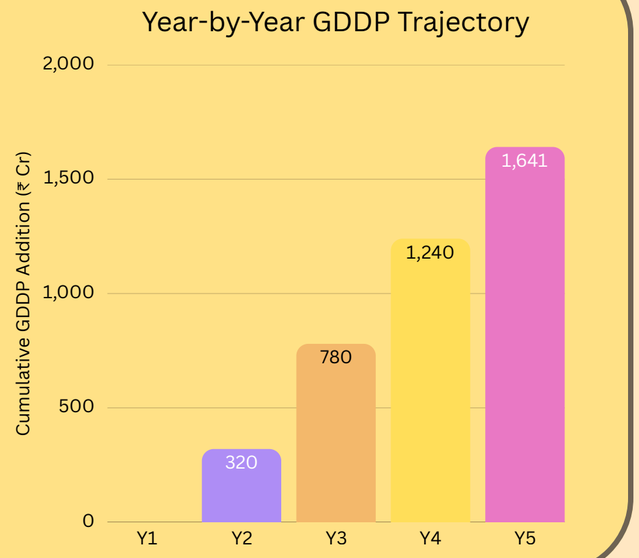
- Tonk products in **state markets**
- District **growth indicator** evaluation
- **Scale** successful models across remaining blocks

Risks	Mitigation
Delayed infrastructure approvals	DPC (District Project Cell) led fast track approval framework
Low credit uptake by farmers/MSMEs	Bank partnership + financial literacy programs
Producer–market disconnect	Multi-channel market integration through mandis, digital platforms and regional trade hubs

A phased strategy to convert infrastructure investment into enterprise growth and market integration

Data Insights Driving the Roadmap

- **34.5%** irrigation coverage → irrigation investment prioritized
- Industry share **18.6%** → MSME cluster development
- **22.5%** district credit gap → credit linkage programs
- Agriculture **42.8%** of GDDP → agro-processing value chains
- Fiscal multiplier **1.85** → Market integration focus



Y1–Y2 builds assets, Y3–Y5 monetises them: 3 lakh Ha irrigated, 3 agro-clusters live, ₹671 Cr credit gap closed via 500 BCs, multiplier 1.85 → 2.04×, ₹1,641 Cr GDDP, ~60,000 livelihoods, net outlay ₹539 Cr after ₹267 Cr convergence leverage.

ALLOCATION SCENARIO COMPARISON: WHY TONKABELL WINS

5 strategies evaluated on GDDP impact, employment, and structural fit over 5 years

₹806 Cr

STRATEGY	ALLOCATION SPLIT	EFF. MULT.	GDDP IMPACT (5YR)	JOBS	VERDICT
TONKABELL OUR PICK Agri + MSMEs + Roads	₹280 · ₹320 · ₹206	**2.04×**	**₹1,641 Cr**	**~60,000**	Optimal ✓
Industry-first MSME heavy, thin agri	₹150 · ₹500 · ₹156	1.98×	₹1,598 Cr	~43,000	No agri base ✗
Agri-first Heavy irrigation focus	₹500 · ₹100 · ₹206	1.77×	₹1,434 Cr	~44,000	Good, not best
Equal split ₹161 Cr across all 5 sectors	₹161 × 5 sectors	1.72×	₹1,388 Cr	~41,000	Diluted impact
Social-first Education + health heavy	₹130 · ₹130 · ₹130 · ₹416 edu	1.44×	₹1,159 Cr	~28,000	15–20yr lag ✗

TONKABELL is the only strategy that sequences correctly: irrigation builds the supply base → MSME clusters capture value → roads + credit unlock market access. Each rupee in one sector amplifies returns in the other two.

THANK YOU

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